

# Connor Magnuson

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## EDUCATION

### University of Illinois Urbana-Champaign

Bachelor of Science in Computer Science

Expected May 2027

GPA: 4.00/4.00, Dean's List

- **Relevant Coursework:** Data Structures, Algorithms and Models of Computation, System Programming, Operating System Design, Computer Architecture, Discrete Structures, Database Systems, Numerical Methods, Calculus I-III, Differential Equations, Probability and Statistics

## PROFESSIONAL EXPERIENCE

### Telematics Software Engineering Intern, Ford Motor Company, Detroit, MI May - August 2026

- Expanded a custom feature flagging system, enabling company-level releases alongside user-level controls
- Developed Ford Pro Telematics features and resolved issues, enhancing **reliability** and expanding **capabilities**

### Android Software Engineering Intern, Midtronics, Willowbrook, IL June - August 2025

- Built a comprehensive **instrumentation testing framework** to validate HMI screenflows across all locales
- Revamped internal **libraries and workflows**, streamlining localization and product customization
- Engineered a **Jenkins CI/CD pipeline** to flag translation display issues and compile screenflow assets

### Software Engineering Intern, RR1 Solutions, Naperville, IL July - August 2024

- Implemented **real-time** server status reporting using WebSockets, enabling monitoring for **thousands of users**
- Utilized protocols and codebases from multiple product teams to deliver **rapid and reliable** client notifications

### Senior Course Assistant, Grainger College of Engineering January 2025 - Present

- Host weekly office hours and **co-teach** in-person recitation sessions for large student cohorts
- Respond to forum posts and provide guidance in C++ and **algorithmic problem-solving**
- Develop lesson materials and create assessment problems to reinforce **computational thinking skills**

## TECHNICAL PROJECTS

### Illini Solar Car Cloud-Based Telemetry System (Java/C++/MQTT/Docker) August 2024 - Present

- Engineered a **redundant, low-latency** cloud telemetry platform to replace a range-limited antenna system
- Streamed **16,000+** CAN messages per minute with synchronized local and cloud datalogging and **>99% uptime**
- Supporting deployment during competitions, enabling **real-time monitoring** of **180+ vehicle metrics**

### Soap Box Derby Live Race Results App (Svelte/AWS) June 2017 - Present

- Co-developing and **scaling** a web app and custom timing platform to broadcast **real-time** race results
- Onboarding and mentoring **15+ race cities**, enabling **thousands of spectators** to follow results remotely

## LEADERSHIP AND TEAM EXPERIENCE

### Illini Solar Car

#### President

May 2026 – Present

- **Mentor** new members and **coordinate project timelines** across multiple subteams and hundreds of students
- Lead weekly board meetings and **facilitate communication** with the university, purchasing office, and donors

#### Strategy and Telemetry Subteam Co-Lead & Board Member

May 2025 – May 2026

- Lead strategy discussions for **system reliability** and telemetry **data ingestion** during competitions
- Support **cross-functional communication** between leadership, software, and hardware teams

### Lead Programmer, First Tech Challenge Robotics Team September 2021 - May 2024

- Developed and optimized **autonomous and remote-control robot** functionality using **innovative** libraries
- Collaborated with mechanical and strategy teams to achieve **two podium finishes** at the Illinois State Finals
- Guided programming team members in iterative testing, debugging, and **embedded state machine design**

## TECHNICAL SKILLS

**Programming Languages:** Verilog, MIPS, C/C++, Java, Kotlin, Python, TypeScript, JavaScript, HTML, CSS

**Embedded & Hardware Systems:** Arduino, CAN, Raspberry Pi, ESP32, MQTT, gdb, Valgrind, Linux

**Full-Stack Development:** React, Svelte, Node.js, Express, REST APIs, WebSockets, Flask, FastAPI, SQL, Firebase, MongoDB, QuestDB

**Cloud & DevOps:** AWS, Azure, DigitalOcean, Docker, Jenkins, Vercel

**Data Analysis & Visualization:** NumPy, Pandas, Matplotlib

**Tools & Other:** Git, GitHub, Bitbucket, Android Studio, VS Code, Jira, LaTeX, Slack, Canva, IFTTT, Microsoft Office